

Supplementary of Motif-FPD: Towards Free-Form Natural Language-Driven Functional Protein Design via Motif-Centric LLM

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1 Overview

This supplementary material provides extended details on the construction, training, and evaluation of the Motif-FPD framework. The content is organized as follows. Specifically, Sec. 2 provides a fully auditable account of dataset construction and quality control for MFPD-210K and MFPD-Bench, covering the train–benchmark split protocol, stage-wise curation statistics, family coverage, query complexity, and expert inspection outcomes. Sec. 3 reports the complete training configuration across all three stages of MFPD-R1, including full hyperparameter tables. Sec. 4 presents ablation results on task ratio configurations for both supervised alignment stages. Sec. 5 conducts an orthogonal cross-tool validation study comprising sequence-level functional consensus, cross-engine structural stability, and case-oriented local structure–function analysis. Sec. 6 gives precise mathematical definitions of the six reward components in the unified reward group. Sec. 7 details the evaluation protocol for all baseline methods, covering the general-purpose LLM baselines, biological expert model baselines, and the unified downstream parsing pipeline. Sec. 8 provides the full prompt templates used for natural language query generation and chain-of-thought annotation in MotifForge.

2 Dataset Construction and Quality Control

The main paper introduces the MotifForge engine and reports the overall scale of MFPD-210K and MFPD-Bench. This section provides a fully auditable account of dataset construction, covering the train–benchmark split protocol (Sec. 2.1), stage-wise curation statistics for the MotifForge pipeline (Sec. 2.2), family-level coverage and long-tail distribution analysis (Sec. 2.3), query complexity characterization (Sec. 2.4), and expert inspection outcomes with inter-annotator agreement statistics (Sec. 2.5).

2.1 Train–Benchmark Split Protocol

A key concern in protein benchmark construction is whether the evaluation split is sufficiently independent from training to support meaningful generalization. We enforce this through sequence-, family-, and topology-level separation. Specifically, benchmark candidates are filtered by MMseqs2 [16] at 40% sequence identity against the full 72,076-fragment pool; at the family level, primary InterPro [4] families are split by deterministic hashing (70% training, 30% benchmark), and overlapping families contribute only benchmark-exclusive functional sub-domains; at the topology level, candidate–training pairs sharing the same CATH [14] level-4 topology are retained only when sequence identity is below 30% and the primary InterPro annotation is benchmark-exclusive.

Table 1: Retrospective sequence-level leakage audit between MFPD-210K and MFPD-Bench. Nearest-neighbor (NN) identity is computed via MMseqs2 all-against-all search against the full MFPD-210K training corpus.

Metric	Value
Benchmark samples with NN identity $\geq 90\%$	146 / 2,905 (5.03%)
Benchmark samples with NN identity $\geq 50\%$	295 / 2,905 (10.15%)
Mean maximum identity to training set	18.36%
Benchmark families absent from training set	219 / 318 (68.87%)
CATH-flagged pairs retained (seq. id. $<30\%$)	63 / 2,905 (2.17%)

Table 2: MotifForge stage-wise statistics. c_s denotes the cumulative retention ratio relative to the raw collection.

Stage	Retained (N_s)	c_s
Raw collection	574,627	100.0%
Cleaning, decomposition & clustering	72,076	12.5%
Task formatting (3 $\times$ expansion)	216,228	–

Each of the 72,076 motif fragments is instantiated across 3 task formats; c_s is not defined for this expansion step.

A retrospective MMseqs2 leakage audit shows that the mean maximum identity between benchmark samples and their nearest training neighbors is 18.36%, with fewer than 5.1% of benchmark instances having a near-identical counterpart at the 90% identity threshold (Table 1). In addition, the 219 benchmark-exclusive families (68.87%) provide substantial out-of-family coverage, indicating that MFPD-Bench is a stringent and substantially leakage-controlled benchmark.

2.2 Data Curation Funnel

Table 2 summarizes the full MotifForge pipeline from raw collection to task-ready data, reporting the stage-wise removal ratio $r_s = (N_{s-1} - N_s)/N_{s-1}$ and cumulative retention ratio $c_s = N_s/N_{\text{raw}}$. Starting from an initial pool of 574,627 sequences (reported as ~575K in the main paper), the pipeline applies BLASTp homology removal at a 60% identity threshold, sequence quality filtering, InterProScan functional decomposition, and MMseqs2 redundancy clustering at 30% identity, collectively reducing the pool to 72,076 high-quality non-redundant motif fragments ($c_s = 12.5\%$). The final task formatting stage instantiates each fragment across three complementary task formats to produce the 216,228-instance MFPD-210K dataset.

2.3 Family Coverage and Long-Tail Distribution

Table 3 reports family-level statistics for MFPD-210K and MFPD-Bench, partitioned into head ($\geq 1.0\%$), mid (0.1%–1.0%), and tail ($< 0.1\%$) frequency tiers. MFPD-Bench exhibits substantially lower head-family concentration and a larger tail-family proportion than

Table 3: Family coverage summary for MFPD-210K and MFPD-Bench. Families are partitioned into head ($\geq 1.0\%$), mid ($0.1\%–1.0\%$), and tail ($< 0.1\%$) frequency tiers. Top-10 share denotes the fraction of samples covered by the ten most frequent families.

Split	#Families	Head	Mid	Tail	Top-10
MFPD-210K	436	60.8%	29.7%	9.5%	38.4%
MFPD-Bench	318	47.9%	34.0%	18.1%	30.1%

Table 4: Query complexity statistics for MFPD-210K and MFPD-Bench. Joint functional+structural rate measures the proportion of queries that simultaneously specify both a functional mechanism and a structural requirement. Multi-hop reasoning cue rate measures the proportion containing compositional or multi-step reasoning cues. High template-overlap rate measures the proportion showing strong similarity to known template patterns.

Metric	MFPD-210K	MFPD-Bench
Avg. tokens/query	27.9	33.6
Median constraints/query	3	4
Joint functional+structural rate	64.2%	73.5%
Multi-hop reasoning cue rate	39.6%	49.1%
High template-overlap rate	7.4%	4.8%

the training corpus, confirming that the benchmark is not dominated by frequent families and is well-suited for evaluating long-tail generalization. The family inventory overlap between the two splits is 90.9% (289/318), with benchmark-exclusive families accounting for 7.2% of benchmark samples.

2.4 Query Complexity Analysis

Table 4 quantifies query complexity along linguistic and semantic dimensions for both splits. Benchmark queries are on average longer and denser than training queries, with a higher proportion of compositional constraints and a markedly lower degree of template overlap, making MFPD-Bench a more demanding test bed for free-form instruction-following in motif-grounded protein design.

2.5 Expert Inspection and Agreement

A total of 1,000 entries were sampled via stratified sampling over family-frequency and query-complexity bins. Three domain experts independently reviewed each entry under blind conditions, assigning binary pass/fail labels on two dimensions: **semantic consistency** and **reasoning chain validity**, with the final label determined by majority vote. The semantic consistency pass rate reaches 995/1000 (99.5%), the reasoning chain validity pass rate reaches 992/1000 (99.2%), and the joint pass rate is 990/1000 (99.0%). Among the 10 jointly rejected entries, 2 failed on semantic consistency only, 5 on reasoning validity only, and 3 on both dimensions.

Inter-annotator agreement is measured via Fleiss' κ (0.51 for semantic consistency, 0.48 for reasoning validity) and pairwise Cohen's κ (0.49–0.53). These moderate values are expected under a heavily skewed label distribution: with a rejection rate below 1%, the chance-agreement baseline P_e approaches 0.98, mechanically

Table 5: Rejection reason statistics on the expert-audited set. Reasons are recorded in a multi-label fashion over the 10 jointly rejected entries; percentages may sum to more than 100%.

Rejection reason	Count	% of rejected	% of all 1,000
Semantic drift from query intent	4	40.0%	0.4%
Functional-site error (motif/position mismatch)	3	30.0%	0.3%
Reasoning leap / missing justification	4	40.0%	0.4%
Sequence implausibility (biophysical inconsistency)	2	20.0%	0.2%
Over-templated expression (insufficient specificity)	1	10.0%	0.1%
Other	1	10.0%	0.1%

suppressing κ . The raw observed agreement across annotators is 99.0% ($P_o = 0.990$), which more directly captures annotator alignment and confirms that disagreements are rare and isolated. Table 5 reports the structured taxonomy of rejection reasons over the 10 jointly rejected entries.

Table 6 reports the detailed difficulty-stratified statistics of both MFPD-210K and MFPD-Bench across all three task formats. Difficulty tiers (Easy, Medium, Hard) are assigned based on the composite difficulty score $\mathcal{D}(s)$ defined in the main paper (Eq. 5).

3 Training Details

All three training stages of MFPD-R1 initialize from Qwen3-8B with the qwen3 chat template, employ LoRA-based parameter-efficient fine-tuning, and share a fixed global random seed of 42. Table 7 consolidates the full hyperparameter configuration across all stages.

Stage-1: Mixed-Task Foundation Adaptation. Training is conducted on 137,387 samples from MFPD-210K in a 40/40/20 task ratio (Fill-Blank Generation, Sequence Selection Generation, Multiple Choice QA). Fill-Blank and Multiple Choice data are provided without CoT trajectories at this stage; Sequence Selection Generation includes reasoning traces. This heterogeneous supervision establishes broad amino acid grammar knowledge and discriminative reasoning capability before intensive CoT training.

Stage-2: CoT-Augmented Instruction Refinement. Stage-2 fine-tunes on 29,045 samples in a 70/20/10 task ratio, initialized from the Stage-1 LoRA adapter. Fill-Blank Generation and Sequence Selection Generation are augmented with complete five-step biochemical CoT trajectories, extending supervision from final-answer prediction to the full reasoning process. The maximum sequence length is extended to 8,192 tokens and the learning rate is reduced to 1×10^{-5} to accommodate longer outputs and fine-grained refinement.

Stage-3: Dual-Core Driven Preference Optimization. This stage is initialized from the Stage-2 adapter, which also serves as the frozen reference policy π_{ref} , distributed across two NVIDIA A100 GPUs. The training corpus consists of 1015 hard samples selected by the confusion-oriented data ranking mechanism. Each prompt generates 4 candidate completions per update step with KL penalty $\beta=0.04$ and sampling temperature 0.8.

4 Ablation Study on Task Ratio Configurations

Table 8 reports the ablation results on task ratio configurations for both Stage-1 and Stage-2. For Stage-1, a balanced ratio of **4:4:2** (Fill-Blank Generation : Sequence Selection Generation : Multiple Choice QA) yields the best overall performance, attaining a Weighted Accuracy (WA) of 67.16 alongside the strongest Fill Blank Thinking

Table 6: Detailed difficulty-stratified statistics of MFPD-210K and MFPD-Bench. Easy, Medium, and Hard tiers are assigned based on the composite difficulty score $\mathcal{D}(s)$ (Eq. 5 in the main paper). Each task format in MFPD-210K contributes exactly 72,076 samples, totaling 216,228. MFPD-Bench totals 2,905 samples.

Dataset	Format	Easy	Medium	Hard	Total
MFPD-210K	Fill-Blank Generation	9,000	42,000	21,076	72,076
	Multiple Choice QA	1,797	34,625	35,654	72,076
	Sequence Selection Generation	1,798	34,626	35,652	72,076
	<i>Total</i>	12,595	111,251	92,382	216,228
MFPD-Bench	Fill-Blank Generation	24	1,510	475	2,009
	Multiple Choice QA	0	145	160	305
	Sequence Selection Generation	0	280	311	591
	<i>Total</i>	24	1,935	946	2,905

Table 7: Complete hyperparameter summary across all three training stages of MFPD-R1. Stage-1: Mixed-Task Foundation Adaptation; Stage-2: CoT-Augmented Instruction Refinement; Stage-3: Dual-Core Driven Preference Optimization.

Hyperparameter	Stage-1	Stage-2	Stage-3
Base model	Qwen3-8B	Qwen3-8B	Qwen3-8B
Random seed	42	42	42
LoRA rank / alpha	128 / 64	128 / 64	64 / 32
LoRA dropout	0.05	0.05	0.05
LoRA target modules	all-linear	all-linear	all-linear
Effective batch size	16	16	8
Max sequence length	4,096	8,192	4,096
Max completion length	–	–	2,048
Num generations	–	–	4
Num epochs	3	5	3
Peak learning rate	1×10^{-4}	1×10^{-5}	1×10^{-6}
LR scheduler	cosine	cosine	cosine
Warmup ratio	0.05	0.03	0.03
KL penalty β	–	–	0.04
Sampling temperature	–	–	0.8
Precision	bf16	bf16	bf16
Gradient checkpointing	✓	✓	✓

Accuracy (FB) of 34.46 and Sequence Generation Accuracy (SG) of 91.64. Increasing the proportion of Fill-Blank tasks beyond this balance (e.g., 7:2:1 or 8:1:1) leads to consistent degradation across all metrics, suggesting that over-concentrating on a single task type undermines the multi-task generalization that Stage-1 is designed to establish.

For Stage-2, the optimal ratio shifts to 7:2:1, where the dominant Fill-Blank with CoT objective drives stronger reasoning-grounded generation, yielding a WA of 73.26 with gains in both FB (42.19) and MC (85.57). This shift is consistent with the curriculum intent: having established broad task coverage via Mixed-Task Foundation Adaptation, the model benefits from concentrated CoT supervision during CoT-Augmented Instruction Refinement to deepen motif-level reasoning. These results confirm that task ratio configurations are non-trivial hyperparameters, and that the optimal balance undergoes a meaningful transition across stages in alignment with the

Table 8: Ablation study on task ratio configurations for Stage-1 and Stage-2. FB: Fill Blank Thinking Accuracy. SG: Sequence Generation Accuracy. MC: Multiple Choice Accuracy. WA: Weighted Accuracy. The optimal ratio used in our final model is highlighted.

Stage	Ratio (FB:SG:MC)	FB ↑	SG ↑	MC ↑	WA ↑
Stage-1	5:2:3	32.17	89.43	74.82	65.47
	5:3:2	31.84	89.91	72.35	64.70
	5:4:1	31.29	89.62	69.14	63.35
	6:1:3	31.96	90.14	75.43	65.84
	6:2:2	32.48	90.37	73.52	65.46
	6:3:1	32.11	90.08	70.17	64.12
	7:1:2	32.73	90.29	74.16	65.73
	7:2:1	32.35	89.87	71.24	64.49
	8:1:1	31.57	89.54	69.83	63.65
	4:4:2	34.46	91.64	75.38	67.16
Stage-2	3:3:4	39.74	91.28	83.94	71.65
	3:4:3	40.31	91.69	82.07	71.36
	3:5:2	40.86	91.94	80.53	71.11
	4:2:4	40.17	91.15	84.31	71.88
	4:3:3	40.62	91.52	82.46	71.53
	4:5:1	39.98	91.44	79.27	70.23
	5:2:3	40.48	91.37	83.12	71.66
	5:3:2	40.24	91.21	81.35	70.93
	5:4:1	39.87	91.05	78.94	69.95
	7:2:1	42.19	92.02	85.57	73.26

progressive learning design of the hierarchical reasoning alignment pipeline, validating the curriculum design.

5 Orthogonal Cross-Tool Validation

To mitigate potential bias from a single annotation and structure-prediction toolchain, we conduct an orthogonal cross-tool validation comprising three components: sequence-level functional consensus across multiple annotation systems (Sec. 5.1), structure-level agreement across two independent folding engines (Sec. 5.2), and a case-oriented local structure–function analysis against native PDB templates (Sec. 5.3). All results in this section are based on computational surrogate assays and should be interpreted as supporting evidence for computationally function-compatible design rather than direct experimental functional validation.

Table 9: Cross-tool functional annotation consensus on 100 sampled instances from MFPD-Bench. Stratified random sampling with a fixed seed preserves the original task-type distribution. InterPro Match@1, Pfam Match@1, CDD Match@1, and Foldseek Match@1 denote the percentage of samples whose top-1 annotation is target-consistent under family-level matching rules. Consensus@2 and Consensus@3 denote the percentage for which at least 2 or 3 tools return target-consistent assignments. Ambiguity denotes conflicting assignments without any target-consistent consensus. Higher is better except Ambiguity.

Method	InterPro Match@1	Pfam Match@1	CDD Match@1	Foldseek Match@1	Consensus@2	Consensus@3	Ambiguity
Motif-FPD (ours)	51.00	51.00	48.00	35.00	58.00	36.00	25.00
GPT-5.4	4.00	3.00	4.00	6.00	5.00	2.00	13.00
Claude-Sonnet-4	2.00	2.00	3.00	6.00	3.00	1.00	36.00
Kimi-K2.5-Thinking	3.00	3.00	4.00	7.00	3.00	1.00	28.00
Deepseek-V3.2-Thinking	1.00	2.00	2.00	4.00	2.00	0.00	33.00
BioT5+ (Expert Baseline)	2.00	0.00	1.00	5.00	3.00	0.00	17.00

Table 10: Structure-level agreement across ESMFold and OpenFold on 50 sampled instances from MFPD-Bench. Values are mean $\pm$ std (sample standard deviation, $n=50$). Global TM-score and Local TM-score measure overall and motif-region structural similarity between the two engines. Global RMSD and Motif RMSD (Å) measure global and motif-centric structural deviation. This experiment evaluates cross-engine consistency, not agreement with experimental ground truth.

Method	Global TM-score $\uparrow$	Global RMSD (Å) $\downarrow$	Motif RMSD (Å) $\downarrow$	Local TM-score $\uparrow$
Motif-FPD (ours)	0.3746 $\pm$ 0.0871	3.7444 $\pm$ 0.9346	3.4418 $\pm$ 0.9649	0.3804 $\pm$ 0.0979
GPT-5.4	0.3789 $\pm$ 0.1535	4.1964 $\pm$ 1.3057	3.7898 $\pm$ 1.2370	0.3762 $\pm$ 0.1563
Claude-Sonnet-4	0.3546 $\pm$ 0.1149	5.0050 $\pm$ 1.2647	4.5260 $\pm$ 1.3156	0.3691 $\pm$ 0.1312
Kimi-K2.5-Thinking	0.3228 $\pm$ 0.0964	4.9060 $\pm$ 1.2115	4.4816 $\pm$ 1.1961	0.3218 $\pm$ 0.1138
DeepSeek-V3.2-Thinking	0.3886 $\pm$ 0.1308	3.6970 $\pm$ 1.2926	3.2416 $\pm$ 1.2765	0.3882 $\pm$ 0.1580
BioT5+ (Expert Baseline)	0.4007 $\pm$ 0.1896	3.9232 $\pm$ 1.0031	3.5084 $\pm$ 1.0683	0.4204 $\pm$ 0.2073

5.1 Sequence-Level Functional Consensus

We conduct cross-tool sequence-level validation using four independent annotation systems: InterPro [4], Pfam [13], CDD [11], and Foldseek [21]. Stratified random sampling with a fixed seed is applied to obtain 100 test instances from MFPD-Bench, preserving the original task-type distribution. The same subset is used across all methods for fair comparison.

As shown in Table 9, Motif-FPD consistently outperforms all baselines across every single-tool criterion and under stricter multi-tool agreement: it attains 58.00 on Consensus@2 and 36.00 on Consensus@3, while all baselines remain in low single digits. The ambiguity rate of 25.00% reflects genuine cross-database vocabulary inconsistencies rather than method failure. We treat this experiment as orthogonal corroborating evidence for the main-paper findings, providing computationally supported indication of function-compatible sequence generation rather than direct functional validation.

5.2 Structure-Level Agreement Across Folding Engines

We evaluate cross-engine structural stability by folding generated sequences with both ESMFold [8] and OpenFold [2] on 50 stratified instances from MFPD-Bench, reporting global TM-score, global RMSD, motif-region RMSD, and local TM-score. This experiment measures cross-engine consistency rather than agreement with experimental reference structures.

As shown in Table 10, our Motif-FPD achieves lower motif RMSD (3.4418 Å) than GPT-5.4 [1], Claude-Sonnet-4 [3], and Kimi-K2.5-Thinking [18], while remaining competitive with DeepSeek-V3.2-Thinking [6] and BioT5+ [15]. Notably, Motif-FPD exhibits the lowest standard deviation across all four metrics, indicating the most consistent cross-engine predictions. BioT5+ and DeepSeek-V3.2-Thinking show marginally better average agreement on certain global measures, though the overall differences are small and the relative ranking varies across metrics. Cross-engine structural stability is therefore not the primary differentiator for Motif-FPD; its principal advantage lies in target-consistent local functional realization under free-form specifications, as further evidenced in Sec. 5.3.

5.3 Local Structure-Function Compatibility

We conduct a two-part case-oriented analysis on the three representative tasks from the main paper (Fig. 5): the ACAD enzyme, the Na/H antiporter, and the ORC3 N-terminal domain. Part I (Table 11) evaluates motif-centric local compatibility using internal reference metrics; Part II (Table 12) evaluates topological similarity and task-specific computational compatibility against natural PDB templates via surrogate assays. Figure 1 provides the corresponding structural visualizations. All scores in Part II are computational proxies and do not constitute direct experimental functional validation.

Part I: Motif-centric local compatibility. For the ACAD case, Motif-FPD is the only method satisfying the functional constraint, achieving the highest pLDDT (74.690), residue recovery (0.500), and perfect spacing validity; DeepSeek-V3.2-Thinking recovers no key residues, while InstructProtein achieves only partial recovery (0.333) with

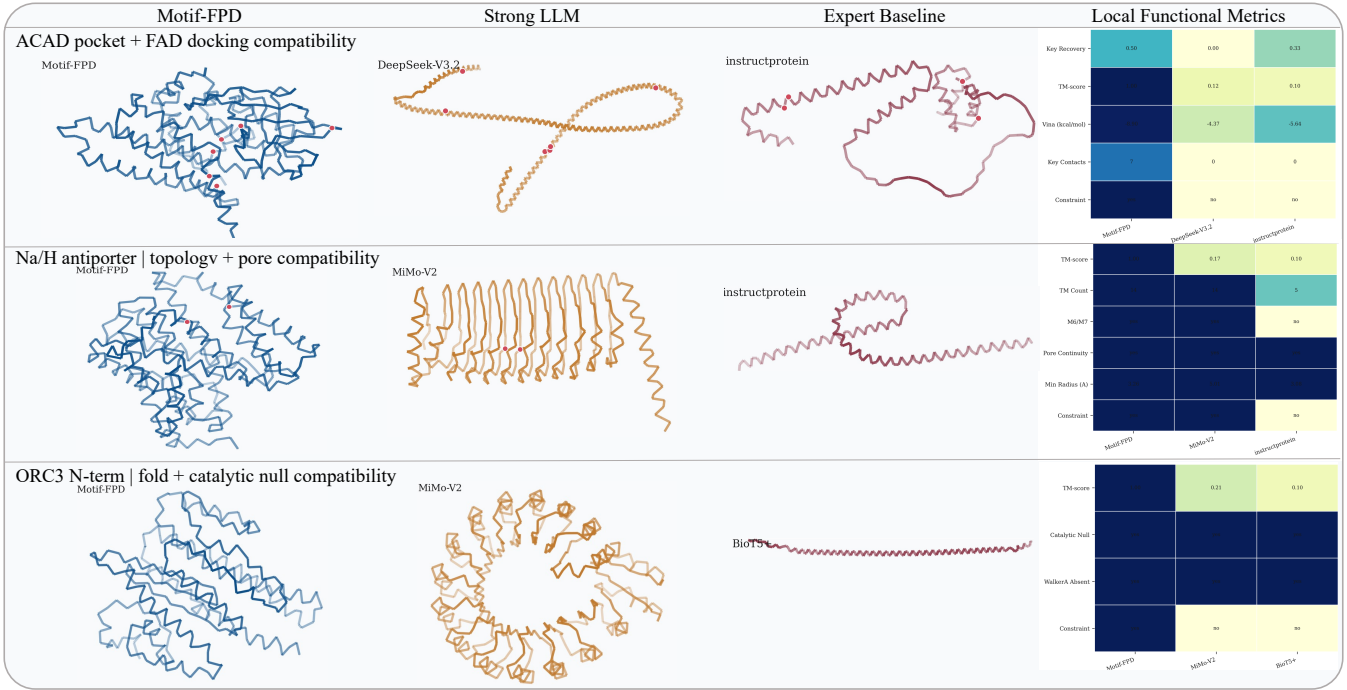


Figure 1: Local structure-function evidence for the three representative NL2Protein cases in the main paper (Fig. 5). Each row corresponds to one case (ACAD enzyme, Na/H antiporter, ORC3 N-terminal domain). Red highlights indicate correctly realized functional domains or conserved motifs required by the user query. Results are shown for Motif-FPD and the compared baselines listed in Tables 11 and 12.

Table 11: Local structure-function metrics for representative cases (Part I: motif-centric compatibility). Key residue recovery: fraction of required residues recovered in the expected local region. Spacing validity: whether the relative order and spacing of key residues satisfy the target motif-level constraints. Local RMSD: computed on the task-relevant region relative to a shared Motif-FPD-centered internal reference used only for cross-method comparison within each case. TMScore (refnorm): normalized local structural similarity under the same internal reference frame; should be interpreted as a relative agreement measure, not an absolute correctness score. Constraint satisfied: whether all task-specific local constraints are jointly met.

Case	Method	pLDDT	Key residue recovery	Spacing validity	Local RMSD (Å)	TMScore (refnorm)	Constraint satisfied
ACAD	Motif-FPD	74.690	0.500	1.00	0.000	1.0000	yes
ACAD	DeepSeek-V3.2-Thinking	57.992	0.000	<u>0.44</u>	63.826	<u>0.1213</u>	no
ACAD	InstructProtein	62.516	<u>0.333</u>	0.00	25.589	0.1028	no
Na/H antiporter	Motif-FPD	87.600	–	1.00	0.000	1.0000	yes
Na/H antiporter	MiMo-V2-Flash	54.465	–	1.00	20.749	<u>0.1698</u>	yes
Na/H antiporter	InstructProtein	<u>67.657</u>	–	0.00	30.217	0.1006	no
ORC3 N-terminal domain	Motif-FPD	28.360	1.000	–	0.000	1.0000	yes
ORC3 N-terminal domain	MiMo-V2-Flash	<u>51.749</u>	<u>0.212</u>	–	<u>17.062</u>	<u>0.2122</u>	no
ORC3 N-terminal domain	BioT5+	71.904	0.100	–	39.960	0.1000	no

zero spacing validity. For the Na/H antiporter, Motif-FPD achieves the highest confidence and perfect spacing validity; MiMo-V2-Flash matches on spacing but shows substantially weaker local structural agreement (local RMSD 20.749 Å vs. 0.000), while InstructProtein fails both. For ORC3, Motif-FPD is the only method jointly satisfying all local requirements including the catalytic-null configuration; BioT5+ attains the highest pLDDT (71.904) but recovers only 10% of key residues, and MiMo-V2-Flash recovers 21.2% with low local similarity (TMScore 0.2122).

Part II: Native template alignment and functional compatibility. Motif-FPD is the only method satisfying all functional constraints

in Part I, and this advantage is corroborated by the Part II surrogate scores. Motif-FPD achieves TM-scores of 0.412, 0.536, and 0.503 against the native PDB templates, exceeding the 0.5 topological similarity threshold for the antiporter and ORC3 cases. For ACAD, the docking score of -8.903 kcal/mol with 7 key contacts exceeds the native ESMFold-folded reference (-8.581 kcal/mol, 8 contacts), confirming strong FAD-binding pocket compatibility. For the Na/H antiporter, the helix-packing score of 0.748 reflects well-preserved transmembrane topology and pore continuity. For ORC3, the AAA+ fold integrity score of 0.779 confirms successful realization of the catalytic-null requirement alongside the required fold topology. All baseline methods produce substantially lower TM-scores and

Table 12: Native template alignment and task-specific computational compatibility scores (Part II). Native PDB template is the closest experimentally determined reference structure retrieved by BLASTp; native rows report reference scores for the native sequence folded under the same ESMFold pipeline, rather than oracle upper-bound controls derived directly from the experimental PDB structure. Global TM-score: TM-align similarity between the ESMFold-predicted structure and the experimental PDB template (>0.5 indicates similar fold topology). Global RMSD to PDB: C α RMSD after TM-align superposition. Functional compatibility score: task-specific computational surrogate assay – estimated FAD-binding Vina docking score (kcal/mol; more negative is better) for ACAD; helix-packing/pore-continuity score (0–1) for Na/H antiporter; AAA+ fold integrity minus catalytic-site penalty (0–1) for ORC3. *Note: functional compatibility scores are task-specific surrogate measures and are not directly comparable across cases or rows; they reflect computationally supported function-compatible design rather than experimentally validated activity.*

Case	Method	Native PDB template	pLDDT $\uparrow$	Global TM-score $\uparrow$	Global RMSD to PDB (Å) $\downarrow$	Functional compatibility score
ACAD	Native seq. folded by ESMFold (PDB ref: 1BUC)	<i>H. sapiens</i> MCAD	84.154	0.2779	14.396	−8.581 kcal/mol (8 contacts)
ACAD	Motif-FPD	–	74.690	0.412	8.743	−8.903 kcal/mol (7 contacts)
ACAD	DeepSeek-V3.2-Thinking	–	57.992	0.183	21.452	−4.369 kcal/mol
ACAD	InstructProtein	–	<u>62.516</u>	<u>0.241</u>	<u>17.836</u>	−5.639 kcal/mol
Na/H antiporter	Native seq. folded by ESMFold (PDB ref: 4CZ8)	<i>E. coli</i> NhaA	94.934	0.3101	13.106	1.000
Na/H antiporter	Motif-FPD	–	87.600	0.536	5.281	0.748
Na/H antiporter	MiMo-V2-Flash	–	54.465	<u>0.487</u>	<u>7.614</u>	<u>0.631</u>
Na/H antiporter	InstructProtein	–	<u>67.657</u>	0.198	24.173	0.214
ORC3 N-terminal domain	Native seq. folded by ESMFold (PDB ref: 5UJO)	<i>S. cerevisiae</i> Orc1 AAA+	80.821	0.1431	25.778	1.000
ORC3 N-terminal domain	Motif-FPD	–	28.360	0.503	6.127	0.779
ORC3 N-terminal domain	MiMo-V2-Flash	–	<u>51.749</u>	<u>0.374</u>	<u>11.284</u>	<u>0.443</u>
ORC3 N-terminal domain	BioT5+	–	71.904	0.289	15.937	0.318

weaker task-specific compatibility scores across all three cases. The lower TM-scores of the native ESMFold rows relative to some designed sequences reflect known ESMFold prediction limitations for certain fold classes (e.g., transmembrane and AAA+ domains), where single-sequence folding without MSA context yields predictions that diverge from the experimental PDB structure; these values therefore serve as pipeline-consistent reference baselines rather than oracle upper bounds, and should not be interpreted as evidence that designed sequences are structurally superior to their natural counterparts.

6 Reward Function Definitions

6.1 Reward Group Definitions

This section provides precise mathematical definitions of the six reward components constituting the unified reward group introduced in the main paper (Eq. 7). All rewards are computed per completion and return a scalar in $[0, 1]$ unless otherwise noted. The weighting factors in Eq. 7 are set as $w_{TQ}=1.0$, $w_{PD}=0.5$, $w_{PF}=1.0$, $w_{PV}=2.0$, $w_{EP}=1.5$, and $w_{IC}=2.0$, reflecting the relative importance of each dimension in the overall optimization objective: biological functional consistency ($\mathcal{R}_{IC}$) and chemical validity ($\mathcal{R}_{PV}$) are assigned the highest weights as they directly determine protein correctness, evolutionary plausibility ($\mathcal{R}_{EP}$) receives an intermediate weight, while reasoning quality ($\mathcal{R}_{TQ}$) and format compliance ($\mathcal{R}_{PF}$) serve as structural constraints with unit weight, and diversity regularization ($\mathcal{R}_{PD}$) carries the lowest weight as a soft penalty term. All weights were fixed after small-scale pilot experiments on a held-out validation subset and were not modified during the final training runs.

Thinking Quality Reward $\mathcal{R}_{TQ}$. $\mathcal{R}_{TQ}$ evaluates the quality of the chain-of-thought derivation enclosed in `<think>...</think>` tags. Any completion whose `<think>` block contains CJK characters is assigned a score of zero to enforce language consistency. For valid

completions, the reward is computed as a weighted sum of three sub-scores, i.e.,

$$\mathcal{R}_{TQ} = 0.55 s_{\text{sec}} + 0.25 s_{\text{char}} + 0.20 s_{\text{len}}, \quad (1)$$

where s_{sec} is the *section coverage score*, defined as the fraction of five required structural headings present in the CoT (*Function & Fold Analysis, Structural Budgeting, Motif Docking, Biochemical Constraints, and Sequence Execution Plan*). s_{char} is the *character legality score*: if any character outside the permitted set (alphanumerics, whitespace, and standard English punctuation) is present the score is 0; otherwise it takes values in $\{0.2, 0.6, 1.0\}$ according to the ratio of alphabetic characters among all non-whitespace characters, with thresholds at 0.40 and 0.55. s_{len} is the *length score*: it equals 1.0 when the word count does not exceed $W_{\text{max}}=200$, and decays linearly as $\max(0, 1 - (w - W_{\text{max}})/W_{\text{max}})$ for $w > W_{\text{max}}$.

Protein Diversity Reward $\mathcal{R}_{PD}$. $\mathcal{R}_{PD}$ penalizes degenerate or low-entropy sequences by measuring trigram ($n=3$) diversity of the extracted amino acid sequence. Let $D = |\mathcal{G}|/|\mathcal{T}|$, where $|\mathcal{G}|$ is the number of distinct trigrams and $|\mathcal{T}|$ is the total trigram count. The reward is defined as:

$$\mathcal{R}_{PD} = \begin{cases} 0 & \text{if } D > 0.8, \\ -\frac{0.8 - D}{0.3} \cdot 0.5 & \text{if } 0.5 < D \leq 0.8, \\ -1.0 & \text{if } D \leq 0.5, \end{cases} \quad (2)$$

where the maximum penalty is -1.0 . Note that $\mathcal{R}_{PD}$ is the only reward component that can take negative values; it therefore acts as a hard regularizer against sequence collapse. The discontinuous jump to -1.0 at $D \leq 0.5$ is intentional: sequences falling below this trigram diversity threshold are considered severely collapsed and warrant a strict hard penalty, as opposed to the soft linear penalty applied in the intermediate range ($0.5 < D \leq 0.8$).

Protein Format Reward $\mathcal{R}_{PF}$. $\mathcal{R}_{PF}$ verifies structural compliance of the output template. A reward of 0.5 is granted if the completion contains a well-formed `<Answer>{ . . . }</Answer>` block, and an additional 0.5 is granted if a `<think>...</think>` block is also present and appears *before* the `<Answer>` block, yielding a maximum score of 1.0:

$$\mathcal{R}_{PF} = 1[\text{Answer present}] \cdot 0.5 + 1[\text{think present} \wedge \text{think precedes Answer}] \cdot 0.5. \quad (3)$$

Protein Validity Reward $\mathcal{R}_{PV}$. $\mathcal{R}_{PV}$ verifies chemical legality of the generated amino acid sequence. Let $\rho = n_{\text{valid}}/n_{\text{total}}$ denote the fraction of residues belonging to the extended amino acid alphabet (20 standard residues plus X, U, O, B). The reward is defined as:

$$\mathcal{R}_{PV} = \begin{cases} 1.0 & \text{if } \rho = 1, \\ 0.5\rho & \text{if } \rho < 1, \\ 0.0 & \text{if no sequence is extracted.} \end{cases} \quad (4)$$

A fully valid sequence receives the maximum score of 1.0, while any invalid residue reduces the reward to at most 0.5, imposing a soft penalty proportional to the contamination rate.

ESM2 Perplexity Reward $\mathcal{R}_{EP}$. $\mathcal{R}_{EP}$ uses the ESM-2 protein language model to assess evolutionary plausibility of the generated sequence. The model is run on CPU in inference mode to avoid memory contention with the policy model during training. The cross-entropy loss $\mathcal{L}$ over masked token prediction is computed, and the perplexity $\text{PPL} = e^{\mathcal{L}}$ is mapped to a discrete reward:

$$\mathcal{R}_{EP} = \begin{cases} 1.0 & \text{if } \text{PPL} < 5, \\ 0.8 & \text{if } 5 \leq \text{PPL} < 10, \\ 0.5 & \text{if } 10 \leq \text{PPL} < 20, \\ 0.2 & \text{if } 20 \leq \text{PPL} < 50, \\ 0.0 & \text{if } \text{PPL} \geq 50. \end{cases} \quad (5)$$

Sequences shorter than 10 residues receive a score of 0, and sequences exceeding 1,024 residues are truncated before evaluation.

InterPro Consistency Reward $\mathcal{R}_{IC}$. $\mathcal{R}_{IC}$ measures functional domain consistency between the generated sequence and the target protein specification. For each training sample, the ground-truth functional domain identifiers are retrieved from a precomputed cache indexed by the protein ID extracted from the input prompt. InterProScan is then executed in real time on the generated sequence to obtain its predicted domain set $\mathcal{F}_{\text{gen}}$, and the reward is the Jaccard similarity between the two domain sets:

$$\mathcal{R}_{IC} = \frac{|\mathcal{F}_{\text{gen}} \cap \mathcal{F}_{\text{gt}}|}{|\mathcal{F}_{\text{gen}} \cup \mathcal{F}_{\text{gt}}|}, \quad (6)$$

where $\mathcal{F}_{\text{gt}}$ is the cached ground-truth domain set. If the protein ID cannot be resolved from the prompt, a neutral score of 0.5 is assigned to avoid penalizing samples with missing annotations. Sequences shorter than 10 residues receive a score of 0. Because InterProScan inference is computationally expensive, $\mathcal{R}_{IC}$ is applied exclusively to the dual-core driven preference optimization stage and is not used during curriculum-like supervised alignment.

ESM3-Keyword — Example Extraction

Input:

Construct a protein sequence containing the homodimerisation region (Qua1) of a STAR domain protein, specifically forming a dimer ...

Extracted keywords ($k=8$):

["homodimerisation", "qua1", "star", "specifically", "forming", "dimer", "region", "domain"]

Conditioning:

Keywords are injected as FunctionAnnotation tokens into ESM3's function track; all other tracks (structure, secondary structure) are left unmasked.

7 Baseline Evaluation Details

7.1 General-Purpose LLM Baselines

For the general-purpose LLM baselines used in the main paper and in the cross-tool validation study, all models are evaluated via their official APIs, with inputs identical to those provided to MFPD-R1 (drawn directly from the original dataset).

Proprietary LLMs include GPT-5.4 [1], Claude-Sonnet-4 [3], Gemini-3.1-pro [5], MiniMax-M2.5 [12], MiMo-V2-Flash [23], and Kimi-K2.5-Thinking [18].

Open-Source LLMs include Deepseek-V3.2-Thinking [9], Hunyuan-2.0-Instruct [20], GLM-4.7 [17], Qwen3.5-Plus [19], Qwen3.5-27B, Qwen3.5-35B-A3B, and Qwen3.5-122B-A10B.

We use a unified prompting protocol that exposes the same instruction and input fields as those used for MFPD-R1. All models are required to produce a single `<Answer>` block containing uppercase one-letter amino acid codes. When supported, decoding uses deterministic or near-deterministic settings (temperature 0.2, top- p 0.95, max_tokens 4096); otherwise, the closest available API-compatible settings are used. All outputs are passed through the same downstream sequence extraction and normalization pipeline described in Sec. 7.3, so that differences in performance reflect genuine model capability rather than model-specific parsing conventions.

7.2 Biological Expert Model Baselines

BioT5+ and InstructProtein (Prompt Reconstruction). Both models [15, 22] receive a unified instruction-following prompt comprising a fixed role header, the full instruction and input fields, and an explicit output-format constraint requiring an `<Answer>` block of uppercase one-letter amino acid codes separated by single spaces. The template is kept identical across both models to eliminate prompt-format confounds.

ESM3-Keyword (Keyword-Conditioned Generation). ESM3 [7] does not support open-ended natural language prompts. We adapt it via a three-step pipeline applied to each concatenated (instruction + input) string. First, standard English stopwords and non-informative tokens are removed via tokenization and stopword filtering. The remaining tokens are then intersected with the ESM3 function-keyword vocabulary to retain only valid annotation terms. Finally, the top $k=8$ keywords ranked by term frequency are injected as FunctionAnnotation conditions for sequence generation.

ProteinDT (Precomputed Subset). ProteinDT [10] results are based on precomputed predictions aligned to test samples by index and input-string matching. Due to inference resource constraints, ProteinDT covers a subset of 1,253 out of 2,600 non-MC samples;

Prompt for Natural Language Query Generation

You are an expert in protein biology and bioinformatics. Your task is to convert the provided protein function description into a **detailed and comprehensive** "Natural Language Requirement (Query)". This Query represents how a knowledgeable expert would express a specific request to construct or find a protein sequence, including its functional mechanisms, structural features, and biological context.

Guidelines:

- (1) **Detailed:** Incorporate specific details from the description, such as mechanism of action (e.g., "catalyzes the hydrolysis of...", "binds to DNA..."), structural features (e.g., "contains a zinc-finger motif", "alpha-helical fold"), biological context (e.g., "involved in stress response", "part of the ribosome"), and interacting partners or substrates if mentioned;
- (2) **Requirement-Oriented:** Start with "Construct a...", "Design a...", "Generate a...", or "Find a protein sequence...";
- (3) **Natural & Fluent:** Use professional biological terminology;
- (4) **No Hallucinations:** Do not invent information not present in the description.

User Input: Please generate a natural language Query for the following protein function description.

Function Description: "{description}"
Return ONLY the Query string.

evaluation is therefore reported on the aligned subset only. All outputs are passed through the same normalization pipeline described in Sec. 7.3.

7.3 Unified Downstream Parsing and Evaluation

For all baselines, a single shared sequence-extraction and normalization pipeline is applied to eliminate evaluation-side confounds. The predicted sequence is resolved via the following fallback order: (1) `parsed_answer` — pre-extracted field if available; (2) `model_response` — parse `<Answer>{ . . . }</Answer>` block; (3) `pred_seq` — direct sequence field; (4) `pred_raw` — raw output, strip all non-amino-acid characters. After extraction, all sequences are normalized to uppercase and filtered to retain only valid amino-acid characters. Validity and all sequence-level metrics are then computed under the same shared evaluation pipeline as Motif-FPD.

8 Prompt Templates

The following two prompt templates are used in the MotifForge engine for natural language query generation and chain-of-thought annotation, respectively. Both prompts are applied to the LLM distillation stage described in Sec. 3.3 of the main paper.

Prompt for Chain-of-Thought (CoT) Generation

Role Definition. You are an expert Structural Biologist and Protein Design Architect. Your task is to perform **High-Precision Reverse Engineering** on a protein sequence.

Input Data. (1) *Target Function:* the natural language description of the protein; (2) *Target Total Length:* total amino acid count; (3) *Reference Sequence Snippet:* the N-terminal start residues only—the full sequence length must not be confused with the snippet length.

Task. Write a Chain-of-Thought (CoT) that acts as a precise architectural blueprint. You must calculate the amino acid "budget" for each structural element and pinpoint where functional motifs should be placed. Do **not** describe what residues exist; instead explain the **design logic**. Be quantitative (use numbers) and topological (reference helix/strand indices).

Anti-Hallucination Constraints. Base all reasoning on the reference snippet; use conditional language ("typically requires", "is commonly found") for uncertain predictions; do not fabricate specific residue numbers unless derivable from the snippet; acknowledge uncertainty when the description is vague;

cross-check structural budgets for biological plausibility; cite conserved motifs in canonical form (e.g., Walker A: G-X-X-G-X-G-K-[T/S]).

Required Reasoning Steps:

- (1) **Function & Fold Analysis:** Identify the most likely fold type; state assumptions; identify the key functional requirement;
- (2) **Structural Budgeting:** Break down the total length into structural components (helices, strands, loops) with explicit residue counts; perform a sanity check that the sum matches total length;
- (3) **Motif Docking:** Specify where functional motifs are likely positioned using topological language (e.g., "the catalytic Lysine should be placed in the loop connecting Beta-strand 1 and Helix 1"); cross-reference with the reference snippet;
- (4) **Biochemical Constraints:** Define hydrophobic core positions, surface charge distribution, and required conserved sequence signatures;
- (5) **Sequence Execution Plan:** Summarize the linear N-to-C arrangement in topological terms (e.g., "N-term → Beta1 → P-loop → ... → C-term");

Output the reasoning **only**, strictly wrapped in `<think>` and `</think>` tags. Do not output the final sequence. Use hedging language ("likely", "typically", "should") for predictions beyond the given data. Limit the thinking section to approximately 600 words.

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